



COSMO INSTRUMENTS (I) PVT. LTD.

SPECIALIST IN AIR LEAK AND FLOW TEST



AIR LEAK TESTER

MODEL LS-R902

ERROR AND TROUBLESHOOTING MANUAL

Error

1.1 When an Error Occurred

The error code is displayed when an error occurs.

The error description, probable causes and the treatments are displayed by tapping the displayed code.

1.2 Error List

Menu to view descriptions, probable causes and treatments and all the errors.

- Go to: Troubleshooting > Error List
- The errors are divided every 10 codes.
- Tapping ↑ ↓ goes forward or back within 10 errors.

Error Messages and Treatments

2.1 ERROR 1: PS Offset Error

Timing: During power-on check procedure

Criteria: Pressure sensor (PS) offset exceeds $\pm 2\%$ of its range.

Probable Cause	Treatment
Test pressure sensor (PS) offset is out of tolerance when the power is turned on.	Adjust the PS offset. Go to: Main > Inspection > Sensor > PS(P1) Contact Cosmo for repair if the offset exceeds $\pm 2\%$ of the sensor range.

2.2 ERROR 2: PS Output Out of Range

Timing: At the end of Pressurization (CHG) or Precharge (PCHG) stage

Criteria: Test pressure exceeds the sensor range in CHG or PCHG stage.

Probable Cause	Treatment
Test pressure sensor (PS) was pressurized exceeding the sensor full-scale.	Adjust the test pressure. Pay extra attention for low pressure models.
Test pressure sensor (PS) offset is out of tolerance.	Adjust the PS offset. Go to: Main > Inspection > Sensor > PS Contact Cosmo for repair if the offset exceeds $\pm 2\%$ of the sensor range.
Cable disconnection or malfunction of the test pressure sensor (PS)	Contact Cosmo for repair.

NOTE

When the error occurred in PCHG, stages only in gray areas are applicable. When the error occurred in CHG stage, the stages in shaded areas with diagonal lines are also applicable. S

2.3 ERROR 3: Improper Test Pressure

Timing:	Test pressure too low:	At the end of Pressurization (CHG)
	Test pressure too high:	Always monitored
	Precharge pressure too low:	At the end of Precharge (PCHG)
	Precharge pressure too high:	Always monitored
Criteria:	Test pressure exceeds upper or lower limit in CHG or PCHG stage.	

Probable Cause	Treatment
"0" is set to the lower limit	Set a numerical number other than "0" to the lower limit.
Upper and Lower limits for Test pressure or Precharge are too close or inappropriate.	Set larger limits. For test pressure limits: Go to: Settings > Advanced Settings > Test Press > Upper Press Limit (TP UL) / Lower Press Limit (TP LL) For Precharge limits: Go to: Settings > Advanced Settings > CHG Options > Precharge Upper Limit / Precharge Lower Limit
Pressurization time is insufficient. (When pressure is lowered.)	Extend CHG timer. Go to: Settings > Advanced Settings > Timer > Pressurization (CHG)
Precharge time is insufficient. (When Precharge pressure is lowered.)	Extend PCHG timer. Go to: Settings > Advanced Settings > CHG Options > Precharge Timer (PCHG)
Fluctuation or a drop in the source pressure	Check the source pressure or the regulator setting. Avoid using air tools branching off the pressure source of the air leak tester. Setting up a dedicated pressure source is recommended.
Leaks from gaskets, part and fittings	Check the gaskets, part and fittings for possible leaks.
Malfunction of the test pressure sensor (PS)	Contact Cosmo for repair.

NOTE

When the error occurred in PCHG, stages only in gray areas are applicable. When the error occurred in CHG stage, the stages in shaded areas with diagonal lines are also applicable.

2.4 ERROR 4: BAL1 Lost Test Pressure

Timing:	At the end of Equalization (BAL1)
Criteria:	Programmed Test pressure lower limit

Probable Cause	Treatment
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Leaks from gaskets, part and fittings	Check the gaskets, part and fittings for possible leaks.
Pilot pressure is not stable, or the regulator is not adjusted properly.	Adjust the pilot pressure between 400 kPa and 700 kPa. Avoid using air tools branching off the pressure source. Setting up a dedicated pressure source for the air leak tester is recommended.
Malfunction of the solenoid valve, SV4, or air-operated valve, AV3.	Contact Cosmo for repair.

2.5 ERROR 5: Leak Limit Settings Error

Timing: At the end of Balance Delay (DL2) At the end of Equalization (BAL1) At the end of Detection (DET)

Criteria: The absolute value of the sum of BAL2 UL and DET UL is greater than the absolute value of the test pressure. The absolute value of the sum of BAL2 LL and DET LL is greater than the absolute value of the test pressure.

Probable Cause	Treatment
The absolute value of the sum of BAL2 UL and DET UL is greater than the absolute value of the test pressure.	Check the Leak Limit. Go to: Settings > Advanced Settings > Leak Limit > BAL2 UL/DET UL
The absolute value of the sum of BAL2 LL and DET LL is greater than the absolute value of the test pressure.	Check the Leak Limit. Go to: Settings > Advanced Settings > Leak Limit > BAL2 LL/DET LL

2.6 ERROR 10: DPS Offset Error

Timing: During power-on check procedure

Criteria: Differential Pressure sensor (DPS) offset exceeds $\pm 30\%$ of its range.

Probable Cause	Treatment
Differential pressure sensor (DPS) offset is out of range when the power is turned on.	Adjust the DPS offset. Go to: Main > Inspection > Sensor > DPS Contact Cosmo for repair if the offset exceeds $\pm 30\%$ of the sensor range.

2.7 ERROR 11: Air Operated Valve Error 1

Timing: At the end of PCHK

Criteria: Test pressure sensor (PS) offset exceeds $\pm 1\%$ of the sensor range.

Probable Cause	Treatment
Pilot pressure is not stable, or the regulator is not adjusted properly.	Adjust the pilot pressure between 400 kPa and 700 kPa. Avoid using air tools branching off the pressure source. Setting up a dedicated pressure source for the air leak tester is recommended.
Test pressure sensor (PS) offset exceeds $\pm 1\%$ of the sensor range.	Adjust the PS offset or enable Auto-Zero feature to reset the pressure residue of the previous test. PS Offset: Go to: Main. > Inspection > Sensor > PS (P1) PS Auto-Zero: Go to: Settings > Advanced Settings > Test Press > PS Auto-Zero > Enable
Charge Delay (DL1) timer is too short	Set the DL1 timer to 0.2 s or longer. Go to: Settings > Advanced Settings > Timer > Charge Delay (DL1)
Malfunction of the test pressure sensor (PS), solenoid valve or air-operated valve.	Contact Cosmo for repair.

2.8 ERROR 12: Air Operated Valve Error 2

Timing: At the end of Pressurization (CHG) or Precharge (PCHG) stage

Criteria: Auto-zero of PS is smaller than 1% of the sensor range at the end of CHG

Probable Cause	Treatment
Pilot pressure is not stable, or the regulator is not adjusted properly.	Adjust the pilot pressure between 400 kPa and 700 kPa. Avoid using air tools branching off the pressure source. Setting up a dedicated pressure source for the air leak tester is recommended.
Pressure source is disconnected.	Check the pressure source and the regulator setting.
Test pressure is too low for high pressure models, H20 and H49.	Adjust the test pressure within the test pressure range.
Malfunction of the test pressure sensor (PS), solenoid valve or air-operated valve.	Contact Cosmo for repair.

NOTE

When the error occurred in PCHG, stages only in gray areas are applicable. When the error occurred in CHG stage, the stages in shaded areas with diagonal lines are also applicable.

2.9 ERROR 14: Air Operated Valve Error 4

Timing: At the end of Air-Blow (BLW)

Criteria: The differential pressure during Air-Blow did not reach the Blow ΔP Limit.

Probable Cause	Treatment
Pilot pressure is not stable, or the regulator is not adjusted properly.	Adjust the pilot pressure between 400 kPa and 700 kPa. Avoid using air tools branching off the pressure source. Setting up a dedicated pressure source for the air leak tester is recommended.
Air-Blow (BLW) timer is too short or Blow Δ P Limit is too high.	Extend the Air-Blow (BLW) timer or lower the Blow Δ P Limit. Air-Blow (BLW) Timer: Go to: Settings > Advanced Settings > Timer > Air-Blow (BLW) Blow Δ P Limit: Go to: Settings > Advanced Settings > Self Check > Blow Δ P Limit
Malfunction of the test pressure sensor (PS), solenoid valve or air-operated valve.	Contact Cosmo for repair.

2.10 ERROR 15: Air Operated Valve Error 5

- Timing:** At the end of Equalization (BAL2). Only for High press and External Press.
- Criteria:** Pressure switch monitoring the pilot pressure for the Balance (BAL) valve is not activated.

Probable Cause	Treatment
Pilot pressure is not stable, or the regulator is not adjusted properly.	Adjust the pilot pressure between 400 kPa and 700 kPa. Avoid using air tools branching off the pressure source. Setting up a dedicated pressure source for the air leak tester is recommended.
Malfunction of the pressure switch monitoring the pilot pressure for the Balance (BAL) valve.	Contact Cosmo for repair. As a provisional measure, the pressure switch monitoring can be disabled. Go to: Settings > Common Settings > Special > PSW Monitoring > Disable

2.11 ERROR 16: Air Operated Valve Error 6

- Timing:** During idle state
- Criteria:** DPS offset exceeded the Idle Δ P Check limit within the programmed Idle Δ P Check Time

Probable Cause	Treatment
DPS offset exceeded its monitoring limit while the air leak tester is in idle state.	Adjust the DPS offset. Go to: Main > Inspection > Sensor > DPS Contact Cosmo for repair if the offset exceeds $\pm 30\%$ of the sensor range.
Exhaust time is insufficient.	Extend Idle Δ P Check Timer or Exhaust timer.

	Idle ΔP Check Timer: Go to: Settings > Advanced Settings > Self Check > Idle ΔP Check Timer Exhaust timer: Go to: Settings > Advanced Settings > Timer > Exhaust (EXH)
Malfunction of the fill valve: SV1 or AV1	Contact Cosmo for repair.

2.12 ERROR 17: Blockage Check Error

Timing: At the end of Pressurization (CHG)

Criteria: The Blockage data exceeded the programmed tolerance sampled Blockage Check data in the Memory.

Probable Cause	Treatment
Something is blocking the air way of the external pneumatic circuits (valves).	Check the external pneumatic circuit (valves).

2.13 ERROR 21: DPS Stopped Oscillating

Timing: Always monitored

Criteria: DPS stopped oscillating.

Probable Cause	Treatment
Malfunction of the DPS or power source or cable disconnection	Contact Cosmo for repair.

2.14 ERROR 22: Stop Valves Closed

Timing: At the end of PCHK (At the end of each stage if closed during measurement)

Criteria: The stop valve monitoring switch is ON/OFF

Probable Cause	Treatment
Stop valves of WORK and MASTER ports are closed, which disturbs the cover from closing. (The stop valve monitoring switch is not pressed.)	Open the stop valves.
If the error occurs even though the stop valves are opened, the stop valve monitoring switch may be malfunctioned.	Contact Cosmo for repair. As a provisional measure, the stop valve monitor can be disabled. Go to: Settings > Common Settings > Special > Stop Valve Monitor > Disable

2.15 ERROR 23: Mastering Error

Timing: At the end of the last iteration of DET for Mastering value sampling

Criteria: Leak data at the end of the last DET iteration exceeded the Mastering Limit in the Mastering Sampling.

Probable Cause	Treatment
Pressurization and stabilization time is insufficient.	Extend Pressurization (CHG) timer and/or Equalization (BAL1) timer. Go to: Settings > Advanced Settings > Timer > Pressurization (CHG) / Equalization(BAL1)
MB1 (Mastering Equalization) timer, MB2 (Mastering Stabilization) timer and/or Mastering Iterations are inappropriate.	Check each setting. Go to: Comp > Mastering Settings > Basic > MB1 Timer / MB2 Timer / Mastering Iterations Ref: The recommended settings for MB1 and MB2 timers are 0.5 s. Make sure that the last DET data is not a negative figure.
Upper and Lower limits for Mastering are inappropriate.	Set larger limits. Go to: Comp > Mastering Settings > Basic > Mastering Comp Upper Limit / Mastering Comp Lower Limit Ref: Typically, the Mastering limits are set to about 120 to 150% of the 1st DET in a Mastering value sampling. Default: ± 250 [Pa]

2.16 ERROR 24: K(Ve) Value Out of Range

Timing: The last DET in K(Ve) Automatic Setup

Criteria: Calculated K(Ve) exceeded 100L.

Probable Cause	Treatment
The current K(Ve) settings does not match the calibrator used for K(Ve) Automatic Setup causing the measured value exceeding 100L.	Check the settings for the calibrator. The items to be set vary depending on the calibrator used for K(Ve) Automatic Setup. ALC: ALC Displacement or ALC Reading Leak Master: LM Flow [mL/min]

2.17 ERROR 25: Leak Limit Out of Range

Timing: The last DET in K(Ve) Automatic Setup

Criteria: K(Ve) Leak limits exceeded the DPS range after K(Ve) Automatic Setup

Probable Cause	Treatment
Leak limits exceeded the DPS range after executing the K(Ve) Automatic Setup.	Change the Leak unit to a pressure unit and perform K(Ve) Automatic Setup again. Go to: Settings > Advanced Settings > Unit > Leak Unit

2.18 ERROR 52 to ERROR 70: System Errors

Usually system errors (ERROR 52 to ERROR 70) are caused by malfunction of electrical components.

System Errors (Timing: At the start of measurement)

Error Code	Description
ERROR 52	AD Communication Error
ERROR 53	I/O Communication Error
ERROR 60	microSD Card Error
ERROR 61	FRAM Checksum Error
ERROR 67	Subnetwork Error
ERROR 68	Fieldbus Error
ERROR 70	TCU Error

Reboot LS-R902 or tap Stop on a measurement screen to cancel the error message. Contact Cosmo for repair after executing System Backup.

Cancelling the Error Message

- 1) Switch the operation mode to Manual (M).
- 2) Tap Stop on a measurement screen to cancel the error message. Rebooting LS-R902 also cancels the error. Or go to: Main > Reboot

2.19 Battery Discharge

When LS-R902 has been used for approx. 10 years, a message window saying "The battery has discharged completely. Please replace the battery and set the date again." is displayed.

When you set the date again, you can use LS-R902. However, since the battery needs to be replaced, contact Cosmo for repair.

3 Large Leak List

The probable cause depends on which stage the large leak judgment is made.

Display	Probable Cause	Treatment
CHG Large Leak WORK side CHG Large Leak MASTER side	There is a large leak on the WORK/MASTER side system.	Check the gaskets, part and fittings for possible leaks.
DL2 Large Leak WORK side DL2 Large Leak MASTER side	There is a large leak on the WORK/MASTER side system.	Check the gaskets, part and fittings for possible leaks.
	Pressurization and stabilization time is insufficient.	Extend Precharge (PCHG), Pressurization (CHG) or Equalization (BAL1) timer. Go to: Settings > Advanced Settings > CHG Options > Precharge Timer (PCHG) Go to: Settings > Advanced Settings > Timer > Pressurization (CHG) / Equalization (BAL1)
BAL2 Large Leak WORK side BAL2 Large Leak MASTER side	There is a large leak on the WORK/MASTER side system.	Check the gaskets, part and fittings for possible leaks.
	Pressurization and stabilization time is insufficient.	Extend Pressurization (CHG) timer and/or Equalization (BAL1) timer.

		Go to: Settings > Advanced Settings > Timer > Pressurization (CHG) / Equalization (BAL1)
DET Large Leak WORK side DET Large Leak MASTER side	There is a large leak on the WORK/MASTER side system.	Check the gaskets, part and fittings for possible leaks.
	Pressurization and stabilization time is insufficient.	Extend Pressurization (CHG) and/or Stabilization (BAL2) timer. Go to: Settings > Advanced Settings > Timer > Pressurization (CHG) / Stabilization (BAL2)

If the problem persists without identifiable causes, please conduct No-Leak Check.

1) Close both the WORK and MASTER stop valves on the rear panel of the tester.

2) Go to: Main > Inspection > Leak Check > No-Leak Check

Contact Cosmo for repair if internal leak is found.

4 Frequent (+) Fails

Follow the procedures below to identify the cause of the frequent fails and remedy the problem.

4.1 Perform a No-Leak Check with the stop valves on the rear panel closed.

If the air leak tester is not leaking, the cause of the frequent fails is something else. Proceed to the next item to check. Contact Cosmo for repair if internal leak is found.

4.2 Check the fixture condition.

Probable Cause	Treatment
Leaks from pipe fittings	Look for leaks in the fittings by performing a bubble test applying soap solution. Redo the piping if needed.
Deformation of pipe	Replace it with pipe that is rigid enough not to deform.

Proceed to the next item to check if the problem persists without identifiable causes.

4.3 Check the sealing condition.

Probable Cause	Treatment
Sealing material is missing.	Place the sealing material.
Sealing surface is contaminated.	Clean the sealing surface.
Sealing material is damaged or worn-out.	Replace it with a new one.
Sealing deforms when the fixture clamps.	Check the following: – Whether there is enough clearance between the sealing material and the groove. – Wear of the stopper. – Whether the size and hardness of the sealing material are appropriate. – If the thrust force of the cylinder has lowered.

Proceed to the next item to check if the problem persists without identifiable causes.

4.4 Check whether there were environmental changes.

Probable Cause	Treatment
Tested part is exposed to the direct wind from air conditioner or fan.	Move the source of the wind to where the wind does not hit the tested parts directly.
Some air tools are branched off the pressure source of the air leak tester causing fluctuation in the pressure source.	Avoid using air tools branching off the pressure source of the air leak tester. Setting up a dedicated pressure source for the air leak tester is recommended.
Air compressor capacity is insufficient.	Use the air compressor whose capacity is large enough.
The current compensation value may not be suitable for the current environmental condition.	Update the compensation value.

Proceed to the next item to check if the problem persists without identifiable causes.

4.5 Check the condition of the tested parts.

Probable Cause	Treatment
Part temperature is higher or lower than the ambient temperature.	Let the part temperature be ambient by adding a cooling/warming buffer in the production line.
The tested parts are wet.	Improve the drying process. Add drying process if there isn't any.
The tested parts get deformed by pressurization.	Add a stopper to prevent the deformation.
Leak due to the gas porosity or internal leak	Look for leaks by performing a bubble test applying soap solution. If no leak is confirmed, there may be internal leaks. If there is a leak, re-evaluate the production process.

5 Frequent (-) Fails

There are two types of the causes for the negative fails. One is caused by a pressure rise in WORK-side circuit and the other is caused by a pressure reduction in the MASTER-side circuit.

Follow the procedures below to identify the cause of the frequent fails and remedy the problem.

5.1 Perform a No-Leak Check with the stop valves on the rear panel closed.

If the air leak tester is not leaking, the cause of the frequent fails is something else. Proceed to the next item to check. Contact Cosmo for repair if internal leak is found.

5.2 By a pressure rise in WORK-side circuit

Probable Cause	Treatment
Sealing is not stable.	Check the following: – Whether there is enough clearance between the sealing material and the groove. – Wear of the stopper. – Whether the size and hardness of the sealing material are appropriate. – Whether the thrust force of the cylinder is too high.
A rise in temperature of the air inside the tested part due to the temperature rise of the cold tested part trying to match the ambient temperature. (Air temperature rise inside the tested part)	Let the part temperature be ambient by adding a cooling/warming buffer in the production line. If the part is wet, add or improve the drying process. Add drying process if there isn't any.

Proceed to the next item to check if the problem persists without identifiable causes.

5.3 By a pressure reduction in MASTER side circuit

Probable Cause	Treatment
There are leaks from the Master or the fittings on the MASTER side.	Check Master part and the fittings for possible leaks by performing a bubble test applying soap solution. Replace the Master part if leak is found. Redo the piping if leak is found in the fittings.
Deformation of MASTER side pipe	Replace it with pipe that is rigid enough not to deform.
Adiabatic compression effect of the Master	The size of the Master Chamber may be wrong, or the BAL2 timer may be too short. Replace the master to the one with good temperature stability. Extend BAL2 timer if possible.

Proceed to the next item to check if the problem persists without identifiable causes.

5.4 By over compensation

Probable Cause	Treatment
The current compensation value may not be suitable for the current environmental condition.	Update the compensation value.